

Heterogeneous Treatment Effects II: Neural Networks

CML Week 6

2026-10-06

Schedule

1. Design-based causal inference & Monte Carlo simulation
2. Covariate adjustment
3. Balancing weights
4. Double robustness, Double ML and TMLE
5. Heterogeneous Treatment Effects
6. **HTE II: Neural Networks**
7. Off-policy evaluation and optimization
8. Experimental design
9. Panel data and the new diff-in-diff
10. Partial Identification
11. Response adaptation
12. Interference



The shift to neural networks

Last time: Tree-based methods (causal forests) and linear approaches (meta-learners)

- Random forests: interpretable, handle interactions naturally
- Meta-learners: flexible frameworks for combining models

Why neural networks?

- High-dimensional, structured covariates (text, images, complex tabular data)
- Can learn flexible representations
- Can incorporate specialized loss functions

The challenge: Standard deep learning is designed for prediction, not causal inference

- Need architectures and objectives tailored to estimating treatment effects
- Theoretical guarantees are critical (not just empirical performance)

Challenges for deep learning in causal inference

Challenge 1: No direct target for supervision

- Supervised learning: observe (X_i, Y_i) and minimize $\mathbb{E}[(Y - f(X))^2]$
- HTE estimation: never observe $\tau_i = Y_i(1) - Y_i(0)$
- Can't directly optimize $\mathbb{E}[(\tau(X) - \hat{\tau}(X))^2]$

Challenge 2: Incomparable representations

- Naive approach: train separate networks $\hat{\mu}_1(\Phi_1(x))$ and $\hat{\mu}_0(\Phi_0(x))$
- Problem: networks learn different representations for treated vs control groups
- When computing $\hat{\tau}(x) = \hat{\mu}_1(\Phi_1(x)) - \hat{\mu}_0(\Phi_0(x))$, we're subtracting predictions from incomparable feature spaces

Challenge 3: Distribution shift

- Treated and control may have different covariate distributions
- Predicting counterfactual $Y_i(0)$ for treated unit requires extrapolation
- Neural networks can extrapolate poorly without explicit constraints



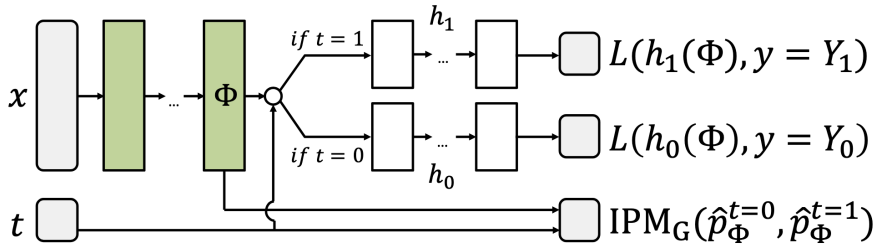
(Shalit, Johansson, and Sontag 2017)

Main contribution: Representation learning for ITE

The solution: Learn a *shared* representation $\Phi(x)$ that balances between:

1. **Predictive information:** $\Phi(x)$ retains information needed to predict $Y(0)$ and $Y(1)$
2. **Balance:** Treated and control distributions look similar in representation space

The idea:



- Same representation $\Phi(x)$ for both treatment arms
- Ensures predictions are made in a comparable feature space
- Estimate: $\hat{\tau}(x) = \hat{\mu}_1(\Phi(x)) - \hat{\mu}_0(\Phi(x))$

Representational balance: If treated and control are balanced in $\Phi(x)$ space, counterfactual predictions are reliable

Balancing representations

Why balance in representation space?

- Original covariate space: treated and control may have very different distributions
- When predicting $Y_i(0)$ for a treated unit, we **extrapolate** to regions with no control
- Models may misbehave when extrapolating

The balancing idea:

- If $P(\Phi(X) \mid A = 1) \approx P(\Phi(X) \mid A = 0)$, then treated and control overlap *in representation space*
- Predictions $\hat{\mu}_0(\Phi(x))$ for treated units are no longer extrapolations

The tradeoff:

- Perfect balance: map everything to the same point $\Phi(x) = c$ for all x
  - Achieves balance but loses all predictive information!
- No balance: use all information $\Phi(x) = x$
  - Preserves information but may require wild extrapolation



Theorem 1: Generalization bound (Setup)

Question: How well does our learned model estimate individual treatment effects?
Setup:

- Representation function: $\Phi : \mathcal{X} \to \mathcal{R}$
- Hypothesis class: $\mathcal{H}$ (functions from representation space to outcomes)
- Learned hypotheses: $h_0, h_1 \in \mathcal{H}$
- ITE estimate: $\hat{\tau}(x) = h_1(\Phi(x)) - h_0(\Phi(x))$

The generalization bound relates three quantities:

1. **Population ITE error:** $\epsilon_{ITE} = \mathbb{E}_{x \sim p}[|Y(1) - Y(0) - \hat{\tau}(x)|]$
   - What we care about but can't directly measure
2. **Empirical factual error:** How well do h_0, h_1 predict observed outcomes?
   - What we can measure during training
3. **Representation distance:** $\text{disc}_{\mathcal{H}}(\Phi) = \sup_{h \in \mathcal{H}} |\mathbb{E}_{p_1}[h(\Phi(X))] - \mathbb{E}_{p_0}[h(\Phi(X))]|$
   - How different treated and control distributions look in Φ space

Aside: Integral Probability Metrics (IPM)

Definition: An IPM measures the distance between two distributions P and Q:

$$\text{IPM}_{\mathcal{F}}(P, Q) = \sup_{f \in \mathcal{F}} \left| \mathbb{E}_{X \sim P}[f(X)] - \mathbb{E}_{X \sim Q}[f(X)] \right|$$

- The choice of function class $\mathcal{F}$ determines the metric
- Larger $\mathcal{F} \implies$ more discriminating distance

Common choices:

- **Maximum Mean Discrepancy (MMD):** $\mathcal{F}$ is the unit ball in a reproducing kernel Hilbert space (RKHS)
  - Computable in closed form using kernel trick
  - $\text{MMD}^2(P, Q) = \mathbb{E}[k(X, X')] - 2\mathbb{E}[k(X, Y)] + \mathbb{E}[k(Y, Y')]$ where $X \sim P, Y \sim Q$
- **Wasserstein distance:** $\mathcal{F}$ is the set of 1-Lipschitz functions
  - 1-Lipschitz: $|f(x) - f(y)| \leq \|x - y\|$ for all x, y (bounded gradient)
  - Stronger metric that accounts for geometry of the space
  - Can be approximated using neural networks with gradient penalty

Theorem 1: Generalization bound (Proof sketch)

The bound:

$$\epsilon_{\text{ITE}}(h_0, h_1, \Phi) \leq \epsilon_0(h_0, \Phi) + \epsilon_1(h_1, \Phi) + \lambda \cdot \text{disc}_{\mathcal{H}}(\Phi)$$

where ϵ_t is the prediction error for treatment group t.

Proof intuition (for a single unit with $A = 1$):

Want to bound: $|Y(1) - Y(0) - \hat{\tau}(x)|$ where $\hat{\tau}(x) = h_1(\Phi(x)) - h_0(\Phi(x))$

$$\begin{aligned} |Y(1) - Y(0) - \hat{\tau}(x)| &= |Y(1) - h_1(\Phi(x)) - (Y(0) - h_0(\Phi(x)))| \\ &\leq \underbrace{|Y(1) - h_1(\Phi(x))|}_{\text{factual error}} + \underbrace{|Y(0) - h_0(\Phi(x))|}_{\text{counterfactual error}} \end{aligned}$$

Key step: How to bound the counterfactual error when we never observe $Y(0)$ for treated units?

- We can bound $\mathbb{E}_{A=0}[|Y(0) - h_0(\Phi(X))|]$ (error on control group)
- The IPM term $\text{disc}_{\mathcal{H}}(\Phi)$ bounds how much this changes when moving from control to treated distribution
- If representations are balanced, counterfactual error $\approx$ factual error on other group

The invertibility of Φ

The condition: Φ must be "invertible"

- Formally: There exists Ψ such that $\Psi(\Phi(x)) = x$ for all x
- Intuition: No information is "thrown away" by the representation
- $\Phi(x_1) = \Phi(x_2)$ only if x_1 and x_2 have the same potential outcomes

Why it's needed:

- Without invertibility: trivial solution $\Phi(x) = c$ for all x achieves perfect balance!
  - » Low IPM, but loses all predictive information about outcomes

Is this realistic?

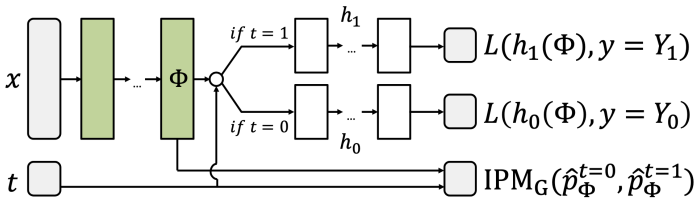
- In practice, we *want* Φ to be a dimensionality reduction
- True invertibility means $\dim(\Phi(x)) \geq \dim(x)$ effectively
- Real implementations: use finite-dimensional representations and hope for "approximate invertibility"
- The theory guarantees don't really hold in practice

Practical takeaway: Need representations that are rich enough to preserve predictive signal, but not so rich that balance is impossible

The architecture

Three components:

1. **Representation network:** $\Phi : \mathcal{X} \to \mathcal{R}$ (shared)
2. **Hypothesis networks:** $h_0, h_1 : \mathcal{R} \to \mathbb{R}$ (predict response surfaces)
3. **Discriminator/IPM network:** $\text{disc}_{\mathcal{H}}(\Phi)$ to enforce balance



Joint loss function:

$$\mathcal{L}(\Phi, h_0, h_1) = \underbrace{\sum_{i:A_i=0} (Y_i - h_0(\Phi(X_i)))^2 + \sum_{i:A_i=1} (Y_i - h_1(\Phi(X_i)))^2}_{\text{Prediction error}} + \alpha \cdot \underbrace{\text{IPM}(P_\Phi^{A=0}, P_\Phi^{A=1})}_{\text{Balance penalty}}$$

- $\alpha > 0$ controls the balance/prediction tradeoff
- Can use MMD, Wasserstein, or adversarial training to compute IPM
- Optimize jointly with respect to Φ, h_0, h_1 (and discriminator parameters if using adversarial approach)

Practical considerations

Strengths:

- **Principled approach:** Theoretical justification for representation learning
- **Flexibility:** Can adapt any neural architecture for Φ, h_0, h_1
- **Explicit balance:** Directly targets the distribution shift problem
- **Influential work:** Spawned a large literature using representation learning

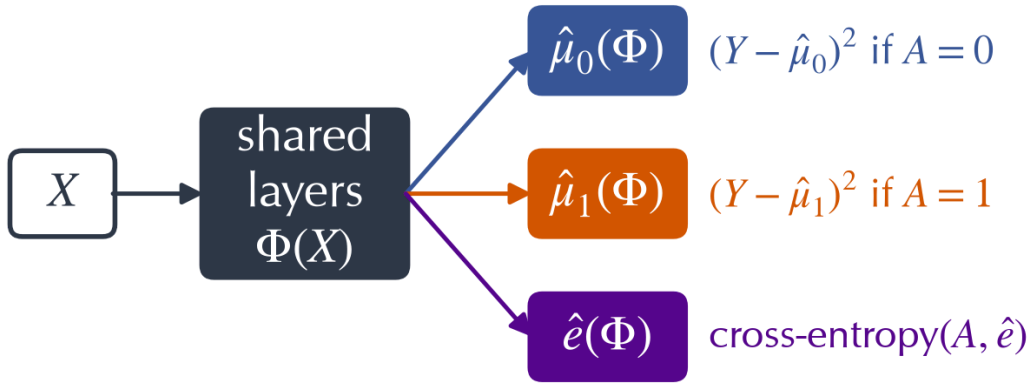
Limitations:

- **Hyperparameter sensitivity:** α tradeoff is critical but problem-specific (no principled way to choose)
- **Invertibility gap:** Practice uses dimensionality reduction
- **Computational cost:** Adversarial training or IPM computation adds overhead
- **Standard causal assumptions:** It isn't magic
- **Bound may not be tight:** May not track the theoretical guarantees

Verdict: A major contribution, but take the bounds seriously but not literally

(Shi, Blei, and Veitch 2019)

Dragonnet architecture



- Shared $\Phi(X)$ as before, plus a **third head** predicting treatment: $\hat{e}(\Phi) \approx e(X)$
- **Why:** adjusting for $e(X)$ alone is enough for identification, so Φ only *needs* the parts of X that predict A. The propensity head pushes Φ to prioritise confounders, at some cost in pure outcome prediction
- Loss: $\mathcal{R}(\theta) = \frac{1}{n}\sum_i \left[\left(Y_i - \hat{\mu}_{A_i}(X_i)\right)^2 + \alpha \cdot \mathrm{CE}\left(A_i, \hat{e}(X_i)\right)\right]$
- No IPM penalty: overlap is handled by estimating $\hat{e}$, not by forcing balance in Φ

Targeted regularisation is TMLE inside the loss

Recall lecture 4: Bang & Robins added $\widehat{e}^{-1}$ as a regressor.

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Dragonnet: a trainable scalar ε on the clever covariate H, trained jointly with θ:

$$\tilde{\mu}_a(x) = \hat{\mu}_a(x) + \varepsilon\, H(a, x), \qquad H(a, x) = \frac{a}{\hat{e}(x)} - \frac{1 - a}{1 - \hat{e}(x)}$$

$$\min_{\theta,\, \varepsilon} \; \mathcal{R}(\theta) + \beta \cdot \tfrac{1}{n} \textstyle\sum_i \left(Y_i - \tilde{\mu}_{A_i}(X_i) \right)^2$$

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Only the second term involves ε, so its first-order condition holds exactly:

$$\textstyle\sum_i H(A_i, X_i)\left(Y_i - \tilde{\mu}_{A_i}(X_i)\right) = 0$$

The plug-in $\frac{1}{n}\sum_i[\tilde{\mu}_1(X_i) - \tilde{\mu}_0(X_i)]$ is then AIPW: the correction term is **zero by construction**.

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› **TMLE** freezes $\hat{\mu}, \hat{e}$, then takes one step in ε. **Dragonnet** targets during training, so Φ moves too

More to read

Paper	Why you'd read it
(Koch et al. 2025)	The review that frames today: a primer on deep learning for causal inference, written for social scientists, with worked tutorials.
(Curth and van der Schaar 2021)	The CATE is often simpler than either potential outcome surface. Architectures that share information between μ_0 and μ_1 can exploit this.
(Ma et al. 2026)	Prior-data fitted networks: pretrain a transformer on simulated causal datasets, then estimate effects in-context on new data.
(Melnychuk et al. 2026)	PFNs carry a prior-induced bias that data never wash out. A one-step posterior correction makes them asymptotically match the efficient ATE estimator.
(Alaa and van der Schaar 2017)	Multi-task Gaussian processes over the potential outcomes, with posterior credible intervals for individual effects.
(Yoon, Jordon, and van der Schaar 2018)	GANITE: a GAN generates the missing counterfactuals, then a second network learns the ITE from the completed data.
(Louizos et al. 2017)	CEVAE: a variational autoencoder that infers latent confounders from noisy proxies.

References

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